EE559 Lab Tutorial 3

Virtuoso Layout Suite XL Editing Introduction

Contents

1 Introduction 2
2 Software Documentation 2
3 Virtuoso Layout Suite XL Editing 3
3.1 Opening the layout editor 3
3.2 Layer Selection Panel 3
3.3 Creating Shapes and Objects 4
3.3.1 Creating Rectangles 4
3.3.2 Creating Wires 4
3.3.3 Creating Pins 4
3.4 Selecting Objects for Edit 5
3.4.1 Selection modes 5
3.4.2 Selecting objects 6
3.5 Editing Objects 6
3.5.1 Moving Objects 6
3.5.2 Copying Objects 7
3.5.3 Deleting Objects 7
3.5.4 Stretching Objects 7
3.5.5 Merging Objects 8
3.6 Saving the Design 8
3.7 Mask Layers 8
3.7.1 Diffusion Areas for Source, Drain, and Substrate Contacts 8
3.7.2 N-well Regions 9
3.7.3 Contacts 9
3.7.4 Metal Layers 9
3.8 Single Transistor Layout 10
4 Layout Verification 12
4.1 Design Rule Check (DRC) 12
4.2 Layout vs. Source/Schematic (LVS) 15
5 Inverter Layout 17
5.1 PCells 17
5.2 Inverter Layout 19
6 Post-Layout Analysis 21
6.1 Parasitic Extraction 21
6.2 Postlayout Simulation 23
7 Hierarchal Layout Editing 25

1 Introduction

The purpose of this lab tutorial is to guide you through the design process in creating a custom IC layout for your CMOS inverter design. The layout represents masks used in wafer fabs to fabricate a die on a silicon wafer, which then eventually are packaged to become integrated circuit chips.

Upon completion of this tutorial, you should be able to:

- Create a mask layout of the CMOS inverter that you have designed earlier
- Check that your layout satisfies the design rules of a 45nm process technology
- Check that your layout passes the automatic verification against the inverter schematic created earlier
- Extract a netlist including parasitic resistances and capacitances from the layout
- Simulate the netlist using Spectre, and compare results to schematic simulations done earlier

- The format of this tutorial is not providing step by step instruction to complete the layout design and verification but it contains enough explanations to help you to finish the basic design work.

- For the evaluation, you need to generate DRC/LVS error free layout of `inverter` and `InverterTest` design and do the layout simulation successfully. When you do the layout simulation, use the input signal pattern that you used in the schematic simulation.

2 Software Documentation

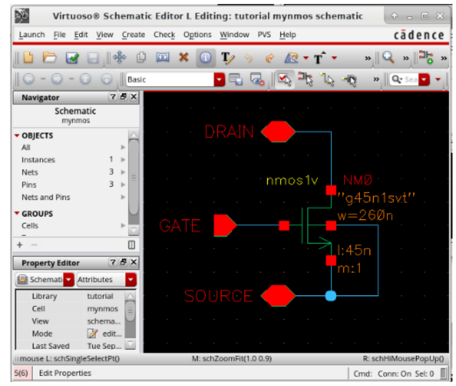
Please refer to the software documentation should you require additional information.

- To access the software documentation, go to **Help - > Virtuoso Documentation Library** in the CIW window.

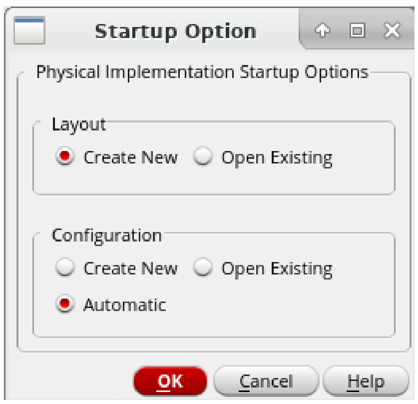
- In the software documentation, more detailed information can be found that you may find helpful.

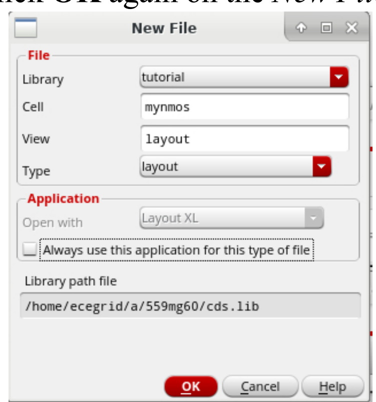
3 Virtuoso Layout Suite XL Editing

3.1 *Opening the layout editor*



- To start up the *Virtuoso Layout Editor*, enter `virtuoso &` in a UNIX window prompt.
- Following the same steps as the previous tutorials, create a new schematic for a cell called *mynmos*
- Place a single *nmos1v* transistor in your design. Set the transistor width to 260nm.
- Create 3 pins called *GATE*, *DRAIN*, and *SOURCE*. Wire the schematic as shown to the right. Check and Save the design.
- From the schematic, select **Launch->Layout XL**. Select *Create New* and *Automatic* then click **OK.** Click **OK** again on the *New File* window.





- A new window titled *Virtuoso Layout Suite XL* will open. This is where you will place the layers to draw the layout of the NMOS transistor.

3.2 *Layer Selection Panel*

The layer selection panel lets you choose the layer on which you create objects (called the entry layer). It also controls which layers are selectable or visible.

- To change the panel to make layers selectable or visible, tick/untick the 'S' or 'V' fields respectively for the particular layer. Invisible layers are automatically set to be unselectable. The text layer color disappears to show the layer is invisible. The layer name turns gray to show the layer is not selectable.

- To make the layers visible, click on the *AV* (All Visible) button. The colored squares showing the layer color reappear, and the shading on the layer name disappears.

- Use the left mouse button to select layers for entry in the layer selection panel. The abbreviation *drw* after each layer name means *drawing*.

3.3 *Creating Shapes and Objects*

We went through the basics of creating rectangles in the previous demo. For completeness, here is a summary of shape and object commands:

3.3.1 Creating Rectangles

- To create rectangles, select a layer (for example, *metal1 / drw*) from the layer selection panel, then select **Create -> Shape -> Rectangle**. In the new window that appears, type the net name you want the rectangle to be associated with (not mandatory). You can choose to leave it blank and name the net later. Hot-key for creating rectangles is '*r*'.

- Note that assigning names to the nets aid in the future layout verification processes, but are not mandatory. However, ensure that the net names on the layout matches the ones in the schematic, otherwise the LVS program (refer to section 4.3) will fail to match the nets.

- Point and click on the first corner of the rectangle, then point to the opposite corner of the rectangle (follow the prompt in the layout window and *CIW*).

3.3.2 Creating Wires

- Another way of creating objects is to create wires. Select a layer from the layer selection panel, then select **Create -> Wiring -> Wire** (hotkey **P**). Press **F3** to open the *Create Wire* menu. In the new window that appears, type the net name you want the polygon to be associated with (not mandatory). You can choose to leave it blank and name the net later or let Virtuoso automatically determine the net name. Set *Snap Mode* to *orthogonal*. The snap mode controls the way segments snap to the drawing grid as you create the polygon by placing its vertices.

- Point and click on the first point of the wire. The *CIW* will prompt for the second point of the wire. Move the cursor to click on a second point. The layout editor will create a wire between the two points.

- Continue to click on a third point that is orthogonal to the path. The layout editor will create a right angle turn in the path to the new point.

- Once you are done creating wire segments, double click to end the wire.

3.3.3 Creating Pins

- In order to perform layout verification after the layout is completed, pins must be created to match the schematic.

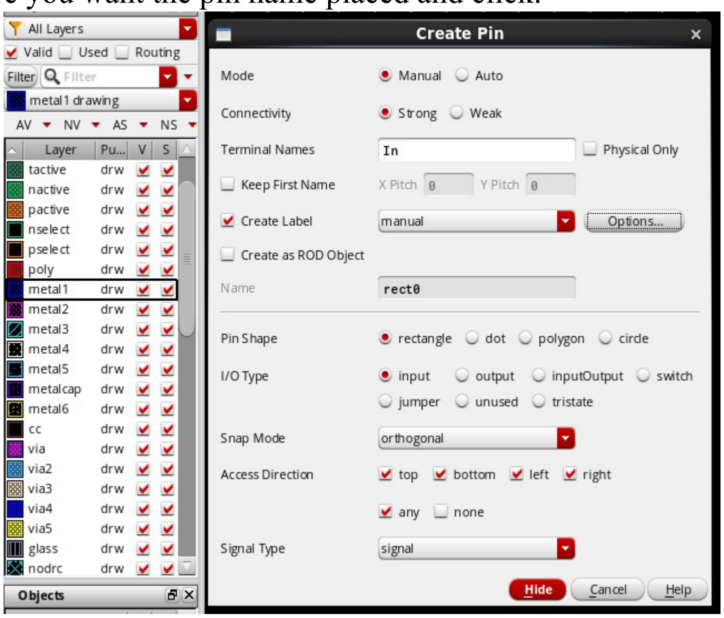
- To create pins, select **Create -› Pin**. In the window that appears, change the *Pin Shape* to *rectangle*.

- Enter the pin name in the *Terminal Names* field. Make sure that the names exactly match the schematic (case sensitive). If you are not sure about the names of the pin nets, open the schematic and check the net properties.

- Turn on the *Create Label* option (required for LVS). Click the **Options** button to change the pin properties. Set Layer Name to "Same As Pin". Set Layer Purpose to "label".
- Select the *I/O Type* accordingly (*input, output, or inputOutput)*.

- Select the layer in the *Layer Selection Panel* (use the layer that has the *drw* abbreviation) and draw the pin in the cellview by clicking on one corner of the pin, followed by the second corner.

- If you have chosen to display the pin name in the cellview, after you have placed the second corner of the pin, the pin name will appear next to cursor. Move the cursor to where you want the pin name placed and click.



3.4 *Selecting Objects for Edit*

3.4.1 Selection modes

To edit an object, first you need to select it. There are two selection modes: full and partial. Press the `F4` key to toggle between selection modes and the mode is displayed in the status banner of the layout window (top).

- In full selection mode (default), you select the entire object when it is clicked. When in full mode, the status banner will display:

`(F)Select:0`

- In partial selection mode, you can select the entire object or just edge or corner of an object. When in partial mode, the status banner will display:

`(P)Select:0`

3.4.2 Selecting objects

- To select an object, set the selection mode and click the object.

- To deselect all objects, click in an empty part of the design.

- To select one or several objects at a time, press the `Shift` key while selecting.

- To deselect one or several objects after they have been selected, press the `Ctrl` key and select.

3.5 Editing Objects

There are several functions that are commonly used to edit objects. They include: move, copy, delete, stretch and merge. Should you require more advanced editing methods, please refer to the *Editing Objects* section in the *Virtuoso Layout Editor User Guide*.

3.5.1 Moving Objects

- To move an object, change to full selection mode and select the object(s). Notice that when you move the cursor within the selected object, the pointer changes to four arrows. This indicates that the object(s) can be moved by clicking and dragging.

- Alternatively, you can choose to select **Edit -› Move** from the drop down menu or use the *Move* icon on the top. The *Move* window appears. After you have selected the object(s), the *CIW* will prompt you for a reference point (start point) for the move. Click on the reference point for the move, and drag the pointer to the destination point. The object will be moved with respect to the reference point.

- Note that in the *Move* window, there is a *Change To Layer* option. This will allow you to move and change the object from one layer to another without having to redraw the object. Check the box to enable the *Change To Layer* function and move the object as usual.

- You can rotate or flip the object (sideways or upside down) by clicking the **Rotate Left/Right** and **Flip Horizontal/Vertical** buttons in the *Move* window before placing the object. You can also do the same by using the right click on the mouse after you have selected the reference point for the move. Shortkey for move function is '*m'*.

3.5.2 Copying Objects

- To copy an object, select **Edit -› Copy** or use the *Copy* icon after you have selected the object(s). After the *copy* window appears, select the object(s) to be copied. The *CIW* will prompt you for a reference point (start point) for the copy. Click on the reference point for the copy, and drag the pointer to the destination point. The object will be copied with respect to the reference point. Shortkey for copy function is '*c'*.

- To copy and paste multiple copies of the object, type in the number of copies in either the *Rows* or *Columns* fields and place the objects in the cellview as usual.

- To copy and paste an array of copied objects, enter both rows and columns. The *CIW* will prompt you to place the first object of the array. After you have placed the first object, continue to place the second column of the array. The distance between the first object and the second will determine the spacing and orientation between the rest of the columns. After you have placed the columns, click to place the rows of the array and complete the array. Similarly, the distance between the first and second rows will determine the spacing and orientation between the rest of the rows.

- Note that in the *Copy* window, there is a *Change To Layer* option. This will allow you to copy and change the object from one layer to another without having to redraw the object. Check the box to enable the *Change To Layer* function and copy the object as usual.

- You can rotate or flip the object (sideways or upside down) by clicking the **Rotate**, **Sideways** and **Upside Down** buttons in the *Copy* window before placing the object. You can also do the same by using the right click on the mouse after you have selected the reference point for the move.

3.5.3 Deleting Objects

- To delete an object, change to full selection mode and select the object(s). Select **Edit -› Delete** or press the `Delete` key.

3.5.4 Stretching Objects

- To stretch an object, switch to partial selection mode and select the object(s) at its corners and edges. Notice that when you move the cursor within the selected object, the pointer changes to an arrow pointing to a line. This indicates that the object(s) can be stretched at the corners or edges by clicking and dragging.

- Alternatively, you can choose to select **Edit -› Stretch** from the drop down menu or use the *Stretch* icon on the left. The *Stretch* window appears. Leave the *Lock Angles* option on unless you need to form nonorthogonal shapes. After you have selected the edge(s) to be stretched, the *CIW* will prompt you for a reference point (start point) for the stretch. Click on the reference point for the stretch, and drag the pointer to the destination point. The object will be stretched with respect to the reference point. Shortkey for stretch function is '*s*'.

3.5.5 Merging Objects

- You can use the merge function to merge two objects of the same layer. To merge objects, select the objects to be merged, then select **Edit -› Merge**.

3.6 Saving the Design

- To save the design, select **Design -› Save** or click the **Save** icon on top.

3.7 Mask Layers

- The mask layers are the various layers shown in the palette and are used to define the location and size of the devices and nets. Each layer can be treated as an individual layer meaning that two different layers have no electrical connection between them even though they happen to overlap.
- The layers are typically in different colors and shading (displayed in the Layer Selection Panel). The colors can be customized and saved to a `display.drf` file.
- The default `display.drf` file can be found in the `/package/eda/cadence/gpdk045/gpdk045_v_6_0/gpdk045/`directory.

3.7.1 Diffusion Areas for Source, Drain, and Substrate Contacts

- Rectangles on the **Oxide** layer are used to define the region where doping is to be applied (except under the polysilicon gate) to form the source and drain of each transistor. For an NMOS transistor, the doping will be n+. For a PMOS transistor, this doping will be p+. It will be shown later how the type of doping is actually specified.
  Rectangles on the **poly** layer are used to define the strips of polysilicon used to form the gate of each transistor and to provide short distance connections between transistors in the inverter.

- The intersection of an **Oxide** and **poly** region defines the channel of a transistor. Since the minimum size of **active** is 120nm and **poly** is 45nm, this means that the minimum transistor width must be 120nm and the minimum length must be 45nm.

- Note that in some cases, it may not be optimal to draw an **Oxide** area as a simple rectangle. The area may have to be one width at the source and drain to accommodate the required clearance around source and drain contacts. It then may need to be notched to obtain the necessary transistor width for the intersection with **poly**.

- The **Oxide** layer is also used to define regions that must be doped to allow a bulk (substrate or well) contact. In p-substrate, the doping must be p+ type. In an N-well (where PMOS transistors are placed), the doping must be n+ type. Note that there are square active layers in the above inverter layout example to form the bulk contacts. Rectangles on the **Nimp** and **Pimp** layers are used to control the type of dopant applied to each diffusion area. Note that these areas must extend past the diffusion area (**Oxide**) by at least 70nm.

3.7.2 N-well Regions

- PMOS transistors must be located in substrate with N type doping. The substrate for the PMOS transistors is formed by diffusing N type dopant into regions of the normally p-type substrate. Rectangles in the **Nwell** layer define these regions in which PMOS transistors can be placed.

3.7.3 Contacts

- 60nm x 60nm squares drawn on the **contact (Cont)** layer will cause metal plugs to be placed into contact with the diffusion areas to form source, drain, and substrate or well contacts.
  60nm x 60nm squares drawn on the **Cont** layer will cause metal plugs to be placed into contact with the **Poly** areas to form poly contacts.

- Metal placed on layer **metal1** will connect with these contacts.

3.7.4 Metal Layers

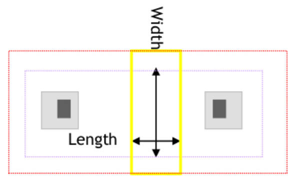
- Rectangles on the **Metal1** layer define regions of metal to be placed in the first metal layer. In this case **Metal1** is used for all inputs and outputs to the inverter.

- In the GPDK045 process, there are several other metal layers available (**Metal2**, **Metal3** and so on). We are not going to use it in this layout since it is not needed. However, in larger more complex layouts, both layers will be needed. Often it is a wise practice to route all signals horizontally on one layer and vertically on another layer.

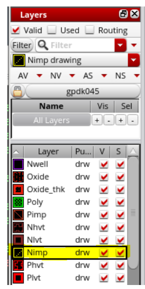
- To connect the **Metal1** layer to the **Metal2** layer, a square on **Via1** is used. To connect the **Metal2** layer to the **Metal3** layer, a square on **Via2** is used.
  You can connect other metal layers together using the appropriate via layers. For example, to connect the **Metal1** layer to the **Metal3** layer, a square on **Via1** and a square on **Via2** is used (with **Metal2** layer in between **Via1** and **Via2**).

3.8 Single Transistor Layout

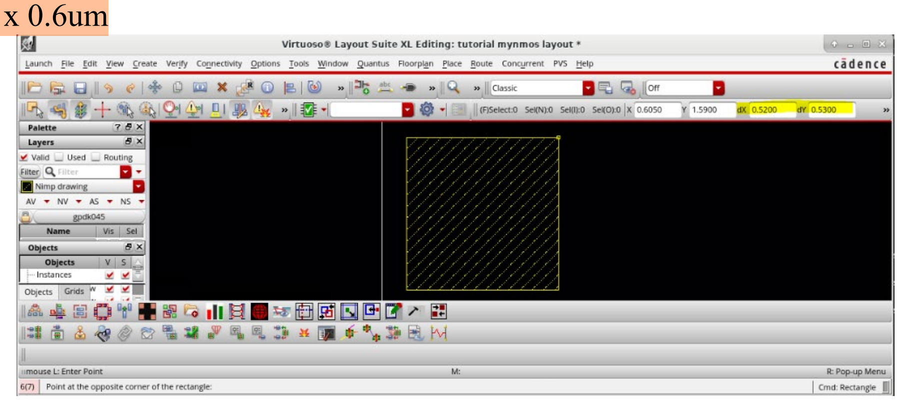
- Open a terminal window and run the command *pdkdoc*. Open "gpdk045_pdk_referenceManual.pdf" in the window that pops up and go to page 44. This will show the layers needed to draw a single transistor



- We need Nimp, Oxide, Poly, Cont, and Metal1 to successfully draw the transistor.
- Return to the *Layout Suite XL* window. Select *Nimp* (N implant) in the Layer Palette

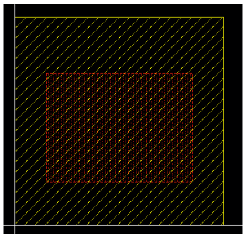


- Select **Create->Shape->Rectangle** (hotkey R). Click once in the center area to start drawing. As you move your mouse around, a yellow box will follow your cursor. In the top right, you should see two fields titled *dX* and *dY*. These are the current size of the box you are drawing in um. Draw a box that is between 0.5um x 0.5um and 0.6um x 0.6um

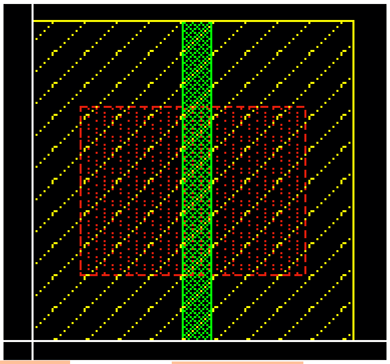


- Click on the box and press **q**. This will open the properties window for the rectangle. You can manually edit the position and size of the rectangle if desired. Click **OK**.
- Press **m** then click on the rectangle. You can move the rectangle Up/Down or Left/Right. You can change the directions that you can move an object by clicking on the snap mode icon or by pressing the **n** key. Move the rectangle so that the lower left corner is close to the origin.
- Select *Oxide* in the layer palette. Press **r** then press **F3**. This will open the *Create Rectangle* form.

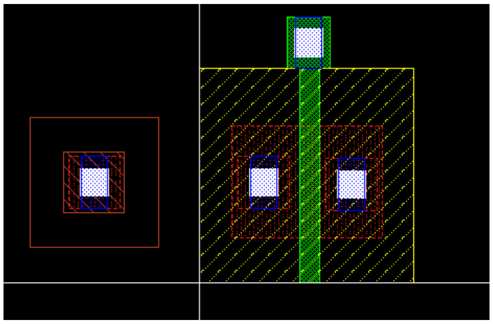
- Check the box for **Specify Size** and set width to 0.35 and height to 0.26. The width can be arbitrarily large, but the height of the Oxide must match the width of the transistor. Place the rectangle approximately in the center of the Nimp rectangle.



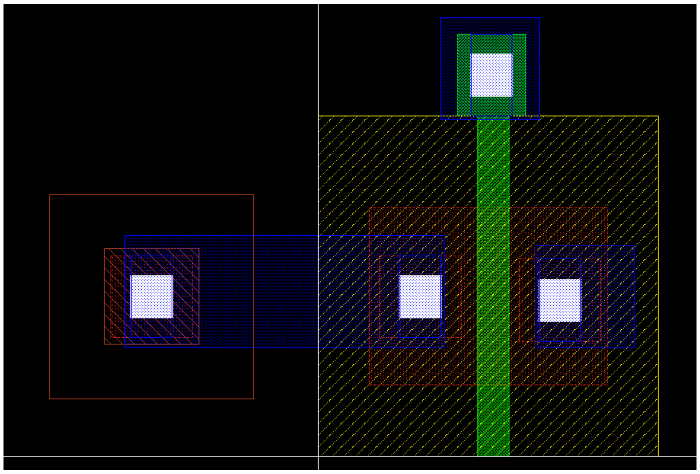
- Select *Poly* in the layer palette. Press **r** then press **F3**. Set the width to 0.045 and set the height to 0.5. The width of the poly defines the length of the transistor, so we set it to 45nm. Place it approximately centered in your two rectangles.



- Select **Create->Via** (hotkey **o**). Select *M1_DIFF* in the *Via Definition* dropdown. Place one via to the left of the poly and another via to the right.
- Select *M1_PO* from the *Via Definition* dropdown. Place the via so that it touches the top of the polysilicon rectangle.
- Select *M1_PSUB* from the *Via Definition* dropdown. Place the via to the side of the whole structure. Make sure it does not overlap any part of the transistor structure.



- Select Metal1 drw from the layer Palette. Press **r** then **F3**. Uncheck the box for specify size. Draw a rectangle that covers one of the transistor terminals (M1_DIFF vias) and the bulk terminal (M1_PSUB via). We also need to draw larger sections of Metal1 due to minimum metal area rules. Draw a metal rectangle around the other two vias that is at least 0.15um x 0.15um.



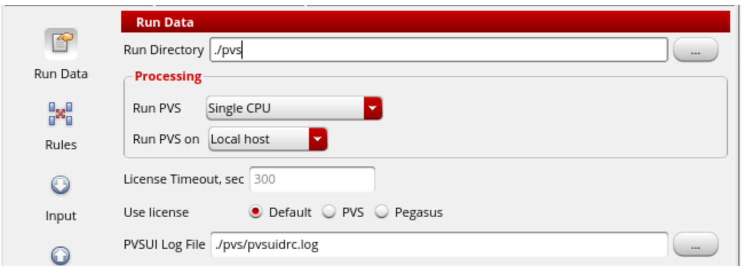
4 Layout Verification

After you have completed the layout, you need to perform several verification procedures to ensure that the layout does not violate any design rules and does actually correspond with the schematic design that you have made earlier.

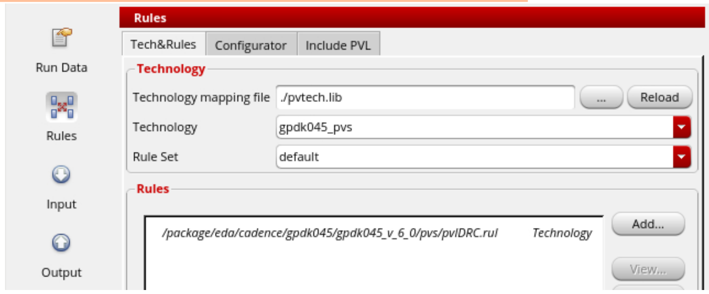
4.1 Design Rule Check (DRC)

- DRC checks your layout against physical design rules defined in the `pvlDRC.rul` file. The rulefile contains all of the rules enumerated in the `gpdk_drc.pdf` file in the `pdkdoc` folder. It will display error information if it finds any part in the layout that violates the design rules. Note that this is only a physical design check and does not verify the actual performance or functionality of the layout design. The file can be found in the `/package/eda/cadence/gpdk045/gpdk045_v_6_0/pvs/` directory.

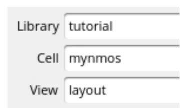
- Next, we will check that our design meets the design rule requirements. Save your current layout. In the *Layout Suite XL* window, select **PVS->Run DRC**.
- Select **Run Data**. Type *./pvs* in the Run Directory Field. This is where PVS will store the DRC results.



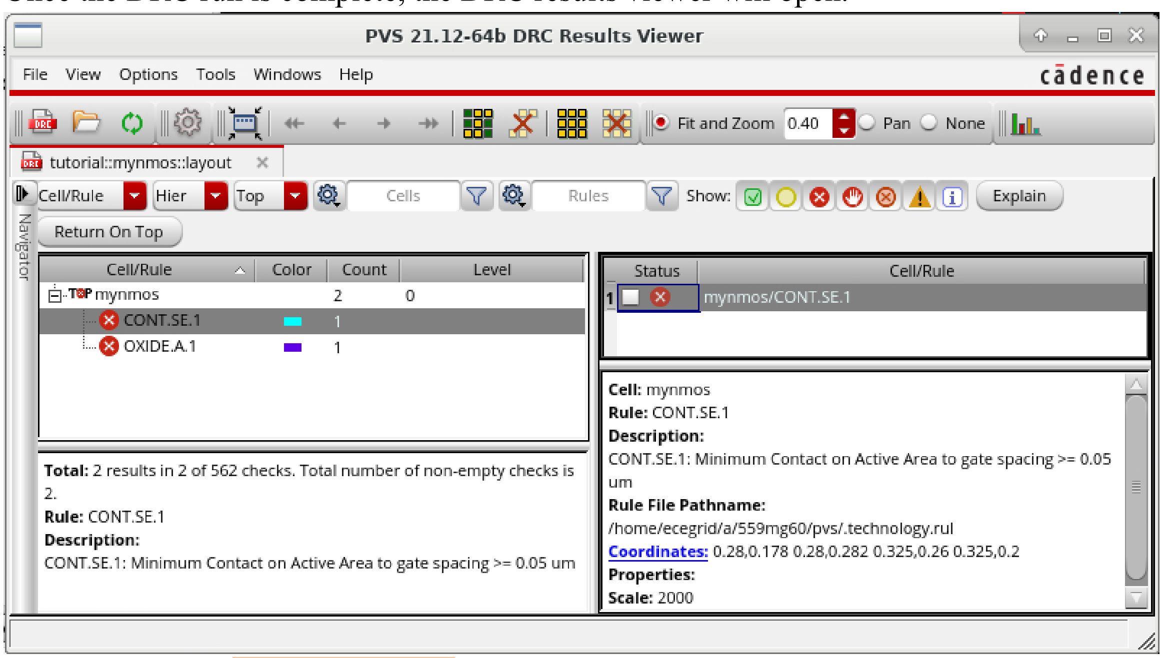
- Select **Rules**. Enter *./pvtech.lib* in the *Technology mapping file* field, then click **Reload**.
- Select *gpdk045_pvs* in the *Technology* dropdown menu.



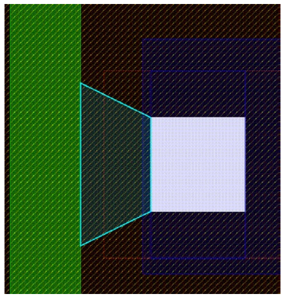
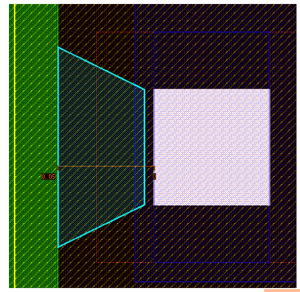
- Select **Input**. Make sure the Library, Cell, and View are correct for your *mynmos* layout.



- Click **Apply** in the bottom left of the PVS window. If you get warnings about overwriting the pvs directory, click **Yes**.
- Once the DRC run is complete, the DRC results viewer will open.

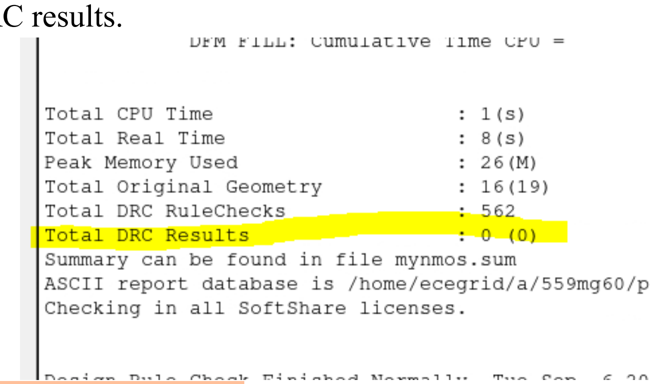


- This example has 2 DRC violations. You may have more or fewer depending on your layout. Click on a DRC violation to see the description. The layout view will also show you the violation.

- In this case, we see that the contact is too close to the poly gate. We need to move it to the right using the move tool (hotkey **m**). To check the dimensions without re-running DRC, we can use the measure tool.
- To measure a distance, press **k**. You can then click on one point then another point to measure a distance. The distance measure tool uses the same rules as the move tool. You can change the horizontal/45°/freeform snapping by pressing **n**. Remove all placed measurements by pressing Shift+k
- The next error is a minimum area violation for the substrate contact. Select the substrate contact via, press **q**, then edit the Oxide Enclosures to match the following:
    - Left: 0.05
    - Right 0.05
    - Top 0.1
    - Bottom 0.1

- Open the *PVS Reports* window and select **ReRun**. Work through the other errors until you have no DRC results.



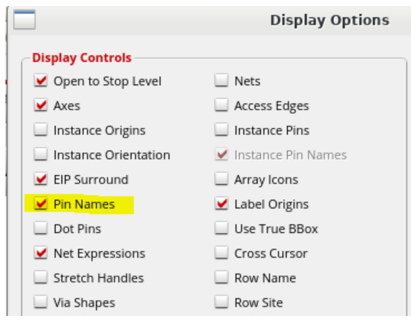
```
DFM FILL: Cumulative Time CPU =

Total CPU Time                    : 1(s)
Total Real Time                   : 8(s)
Peak Memory Used                  : 26(M)
Total Original Geometry           : 16(19)
Total DRC RuleChecks              : 562
Total DRC Results                 : 0 (0)
Summary can be found in file mynmos.sum
ASCII report database is /home/ecegrid/a/559mg60/p
Checking in all SoftShare licenses.
```

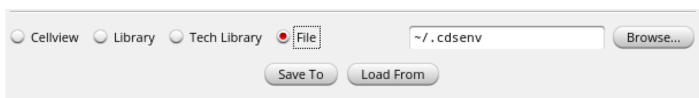
- When Total DRC Results is 0, your design has satisfied all manufacturing requirements.
- It is also possible to run a DRC on a specific area. To do this, select **Input** in the *PVS DRC Run Submission Form* then check the box *Area to Check on Layout* then select *Select on Viewer*. Select the area on the cellview that you want a DRC to be performed on by clicking on the first point of the rectangle followed by the second point. The coordinates of the points will be entered.

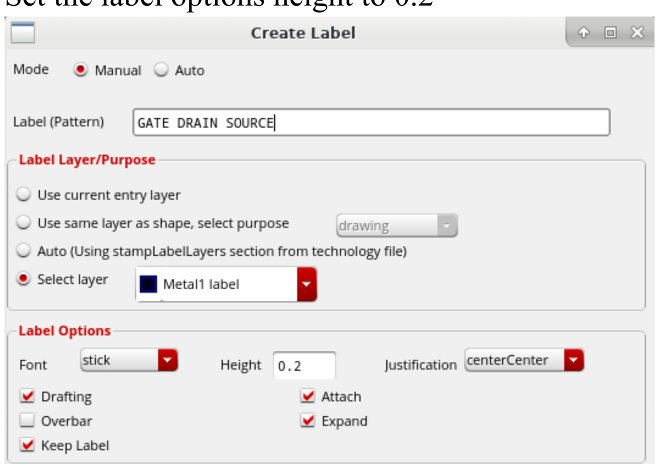
4.2 *Layout vs. Source/Schematic (LVS)*

- Next, we need to check Layout vs Source (LVS). Close all DRC windows before proceeding. One item required for LVS that is not required for DRC is pin labels. This tells the LVS engine what pieces of metal correspond to the pins in the schematic.

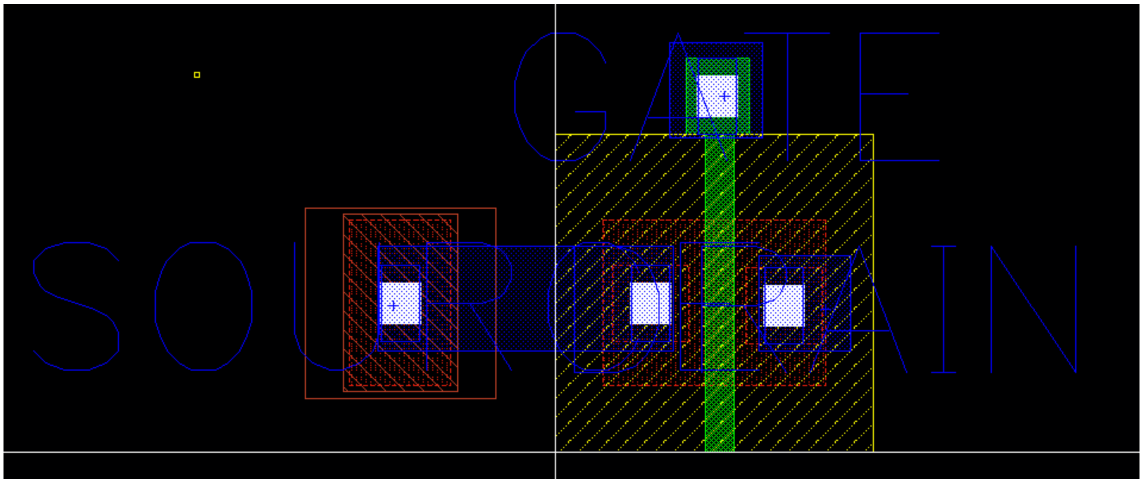
- In the *Layout Suite XL* window, press **e** then check the box for “Pin Names”. At the bottom, select **File** then press **Save To**. This will save your display options preferences. Press **OK**



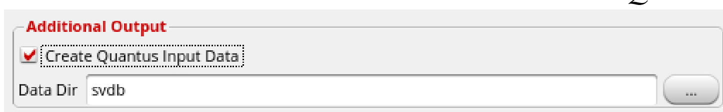


- Select **Create->Label.**
  - Enter “GATE DRAIN SOURCE” into the label field.
  - Set the layer to **Select Layer**: *Metal1 label*
  - Set the label options height to 0.2
  - 

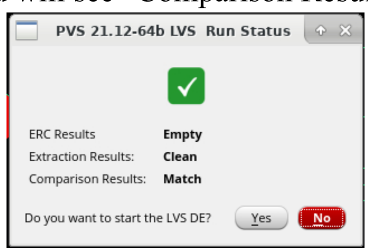
- Click once on the metal1 gate connection (M1_PO via). If prompted, select Metal1 then OK. Repeat for the drain (right side) and source (left side also connected to the substrate via). Save your layout.



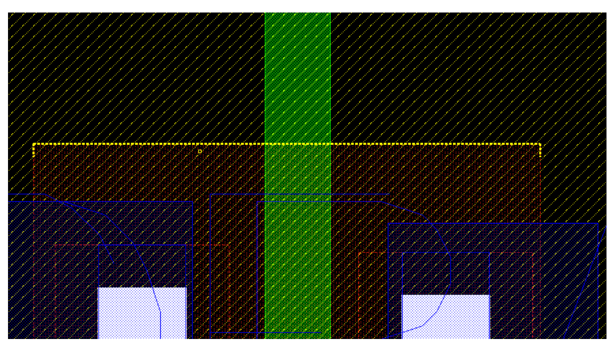
- Select **PVS->Run LVS**. Set up the form using the same steps as DRC (Set the Run Data, Rules, and Input sections).
- Select **Output**. Scroll to the bottom and check the box *Create Quantus Input Data*



- Click **Submit**
- If everything went well, you will see "Comparison Results: Match".

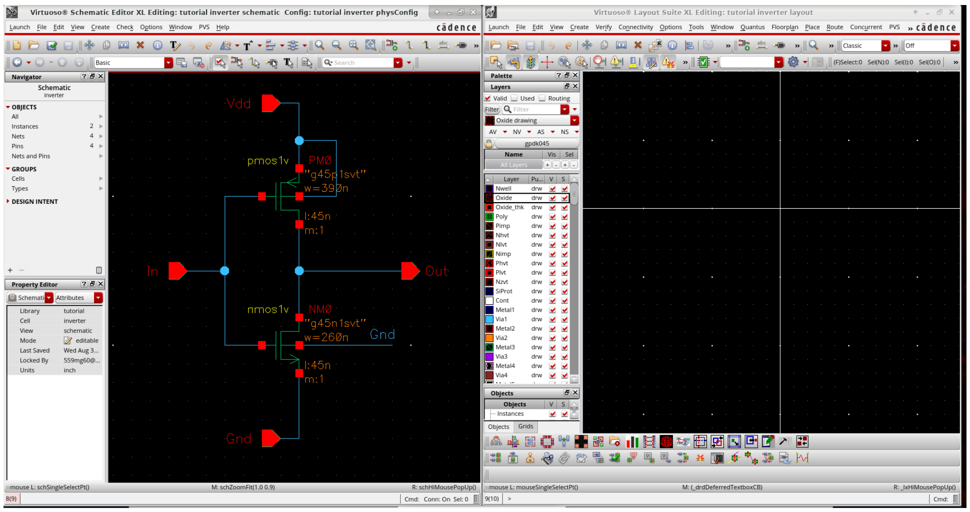


- Click **Yes** to open the LVS DE (Debug Environment). This is where you would see any errors if your schematic and layout do not match.
- Select **Tools->Graphical LVS Debugger**. This is the tool you would use to identify what portions of the layout and schematic do not match.
- Return to the layout. Press **s** and hover over the top of the Oxide rectangle. You should see the edge highlight itself. Click on the edge and drag it up to stretch the rectangle.
- Re-run LVS. The layout and schematic no longer match as we have made the transistor too wide. Explore the Debug Environment to see how you could debug the mismatch.



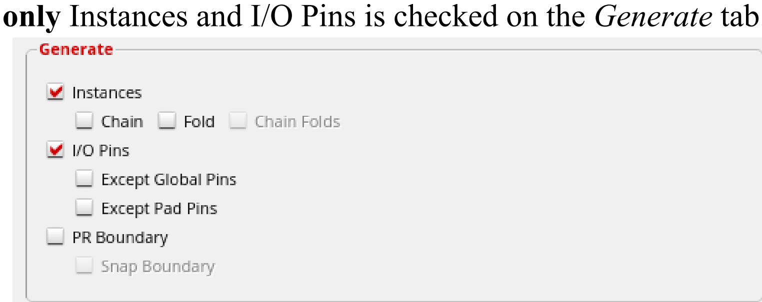
5 Inverter Layout

- Close all the windows you have open.
- Use the Library Manager open the cell *inverter* view *schematic*. In the *Schematic Editor L* window that appears, select **Launch->Layout XL.**
- Choose *Create New* for layout and *Automatic* for configuration. Select **OK** twice. You should have the *Virtuoso Schematic Editor XL* and *Virtuoso Layout Suite XL* editors shown side by side.

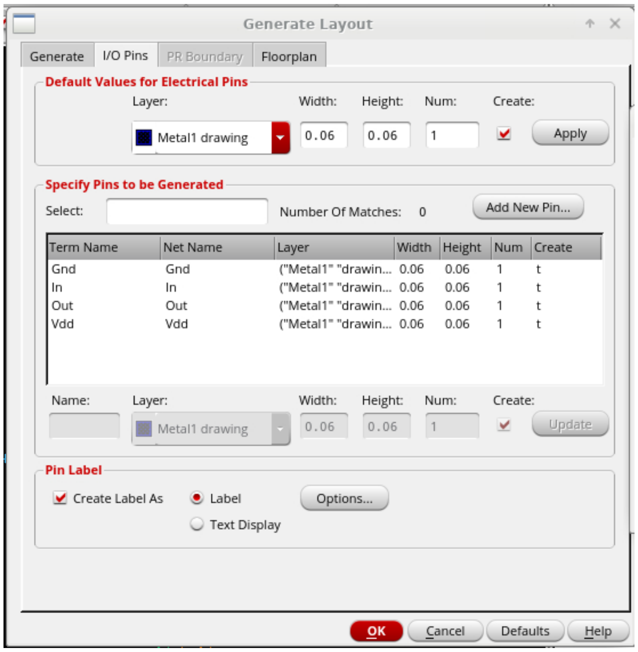


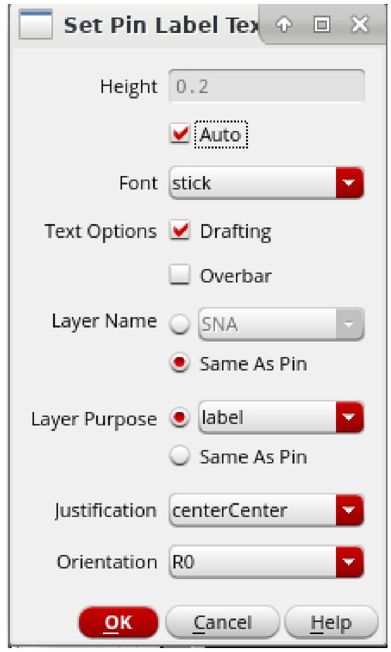
5.1 *PCells*

- Rather than hand draw every transistor, we can use Pcells to generate DRC correct layouts automatically. This also creates transistors that are automatically linked to the schematic transistors.
- Select **Connectivity->Generate->All from Source.** The *Generate Layout* window will open.
- Ensure that **only** Instances and I/O Pins is checked on the *Generate* tab.

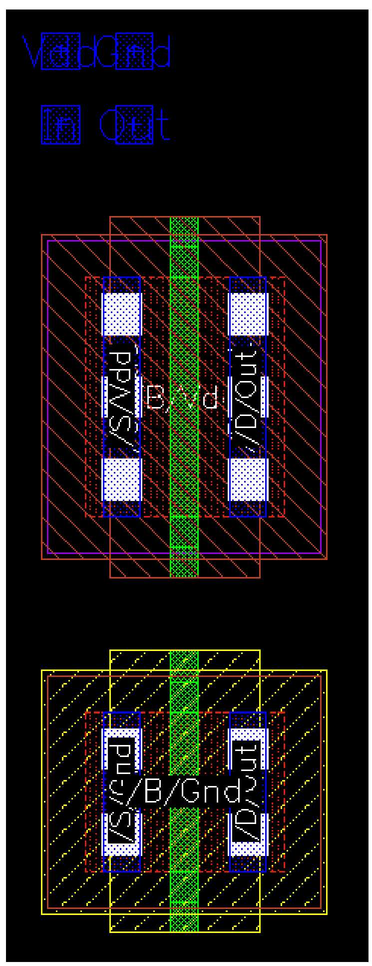


- Switch to the *I/O Pins* tab. On this tab, we can customize how the I/O pins are generated. Check the box for **Create Label As**, mark the radio button for **Label**, then select **Options…**
- In the *Set Pin Label* window, check the box for *Auto*, set the layer name to *Same as Pin*, and set the layer purpose to *label*.





- Select **OK** on both open windows. This will generate all of the components for your inverter layout.



- Note that the automatically generated layouts do not include substrate (Psub) or Nwell connections. Click on either transistor and press **q** to open the properties window. Find the dropdown for *Bodytie Type*. You can create an *Integrated* bodytie if the source and bulk should be tied together. If the source should be connected to a different node than the bulk, then you can select the *Detached* bodytie. Investigate all of the options for the bodyties.
- When doing more complicated layouts, not every transistor needs its own bodytie. In the GPDK45 technology, you must have a bodytie within 20um in the same well. For NMOS, this means anywhere within 20um of the device. For PMOS, the bodytie must be in the same Nwell area.
- Once you have explored the bodytie types, set both the NMOS and PMOS to Integrated Left Tap bodyties.
- Note that in the PCell, you can also add an automatically generated gate connection and change the metal width on the source and drain. Adding a gate connection to the top of the NMOS or bottom of the PMOS will be useful for the inverter layout.

5.2 Inverter Layout

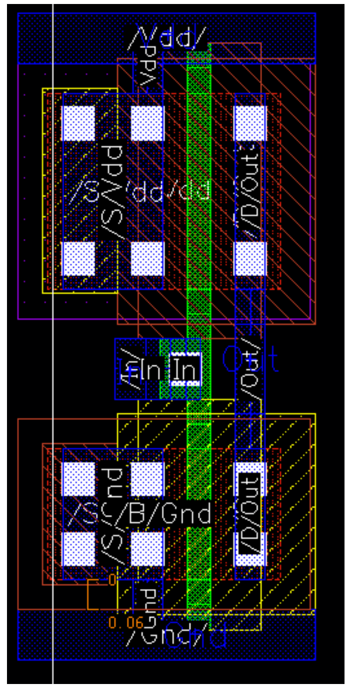
The pictures in this section present an inverter layout very similar to the one you are about to create.

This layout is in the style of standard cells used for automated placement and routing of random logic. This does not, however, mean that this style of layout is bad for custom layout. It has some very useful features. In particular,

- It is designed so that multiple instances of the cell can be connected together by abutment (i.e., placed immediately to the left and right of each other). The power, ground, input, and output connections line up and will be connected. Of course, you may wish to have the input and output not line up so that you can have the power and ground connections connect up without necessarily connecting the input and output together.

- The layout lends itself to a left to right signal flow in the metal layer (used for the input and output) as well as vertical signal flow for short distances in polysilicon.

- If other types of logic cells have the same layout spacing between power and ground, then cells of various types can be chained together easily.



- The final layout should look like as in the above figure.

- You will need to move (**m**) and stretch (**s**) the autogenerated pins to the correct locations.

- Complete your layout to look like the above layout. It doesn't need to match perfectly.

- You should use the **Create Wire (p)** tool to connect different parts of the design.

- You may find setting **DRD Notify** to on (click the  icon) makes generating a DRC clean layout easier. Note that this **is not** a replacement for running DRC.

- Selecting **Connectivity->Check->Against Source** can be a useful tool for identifying potential LVS issues ahead of time. Note that this **is not** a replacement for running LVS.

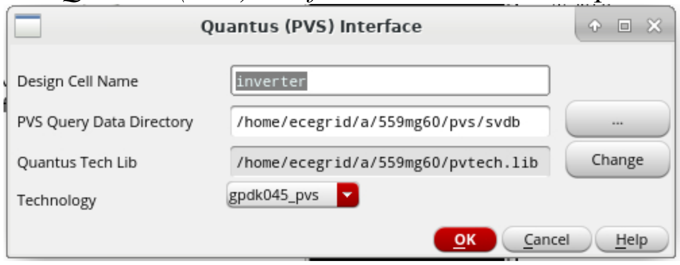
- Use the same procedure for DRC and LVS as the single transistor layout.

6 Post-Layout Analysis

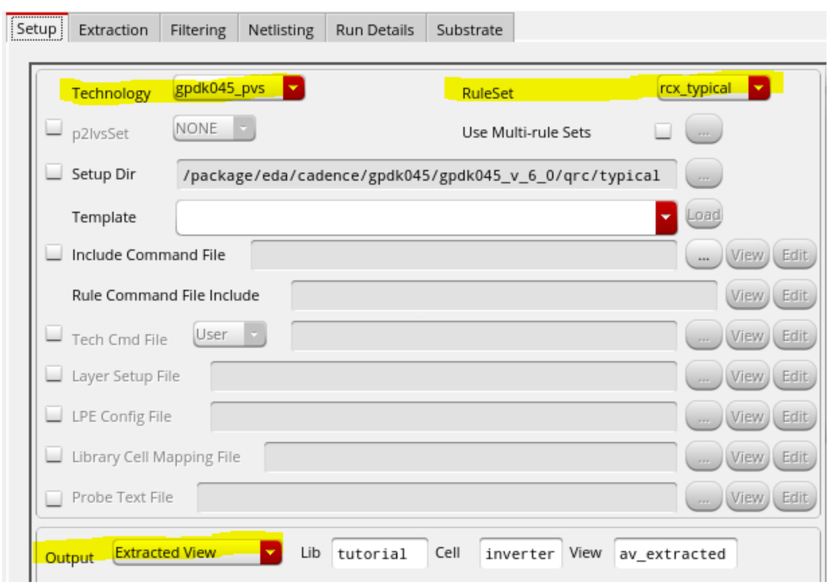
After you have verified that the layout matches the schematic, we can use Quantus to extract the parasitics introduced in the wiring of the inverter. The main difference between a schematic simulation and a layout simulation is that in a layout simulation, parasitic resistances and capacitances are extracted from the layout based on the physical dimensions and are used in the simulation; therefore it will provide a more realistic simulation of the performance of the fabricated device.

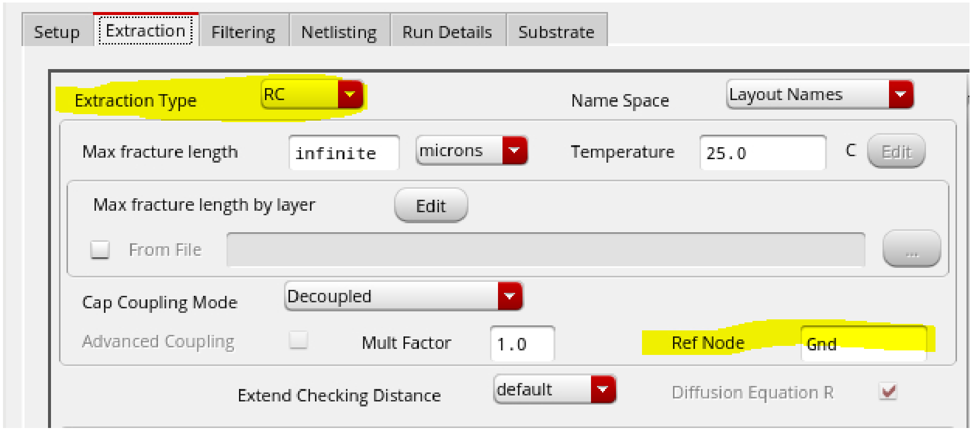
6.1 Parasitic Extraction

- In your layout, select **Quantus -> Run PVS – Quantus**
- The *Quantus (PVS) Inteface* window should open and have the parameters pre-configured

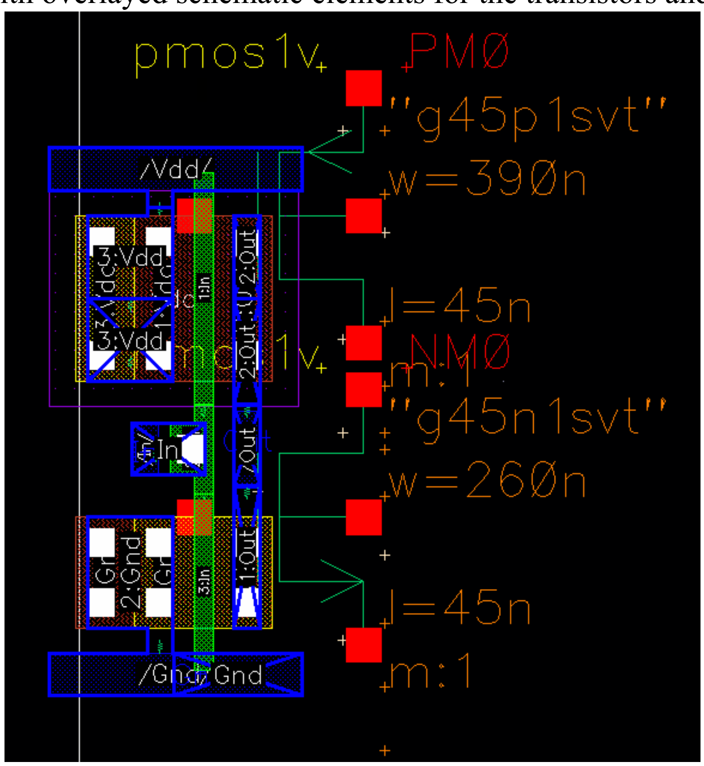


- Select **OK.** The *Quantus (PVS) Parasitic Extraction Run Form* will open.
- In the *RuleSet* dropdown, select *rcx_typical*
- In the *Output* dropdown, select *Extracted View*
- Change to the *Extraction* tab. In the *Extraction Type* dropdown, select *RC.*
- In the *Ref Node* text box, enter the name of the ground net (Gnd). If your design has a different name for the ground net, enter it here instead.



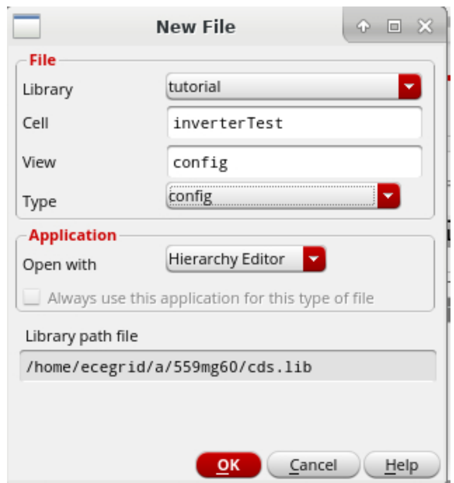


- Click **OK** to start the parasitic extraction.
- Open the library manager and open the newly created *av_extracted* view. You should see your layout with overlayed schematic elements for the transistors and parasitics.

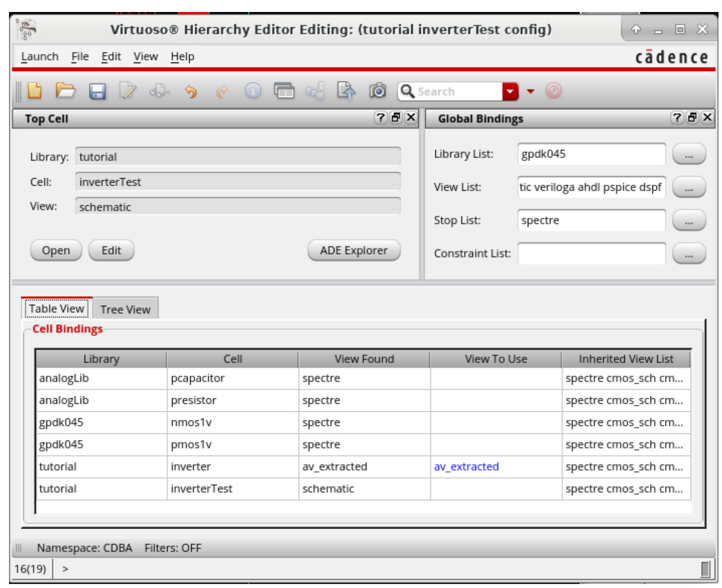


6.2 *Postlayout Simulation*

- Open the schematic for inverterTest. Replace the *inverterp* cells with *inverter* cells. Remove the ideal capacitors.
- Open up the *Analog Design Environment* window by selecting **Launch -> ADE Explorer**. Ensure that you have your stimuli set up correctly (or you have the simulation configured to use a vector file).
- Return to the *Library Manager*. Create a new cellview in the inverterTest cell called *config* with the type set to *config*. The config view is used to specify the simulation hierarchy of your design. It will tell the simulator what views to use in order to netlist the design for simulation.

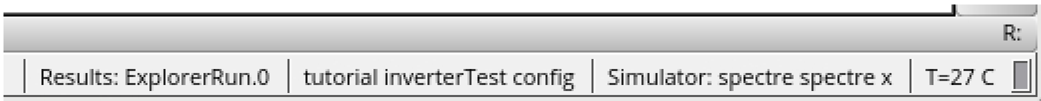


- In the *New Configuration* window that opens, click **Use Template** and select *Spectre* from the dropdown.
- Select *schematic* in the *Top Cell -> View* section.
- Replace *myLib* with *gpdk045* in the *Global Bindings -> Library List* section. Press **OK**
- A window that lists the Library, Cell, and View of every element in your design will open. Right click on the row for *inverter* then choose **Set Cell View->av_extracted**. Save the config view.
- Note that ideal resistors and capacitors have been added from analogLib. These devices represent the parasitics in the layout.
- You can also go to the *Tree View* tab to set only some of the inverters in *inverterTest* to the parasitic devices.

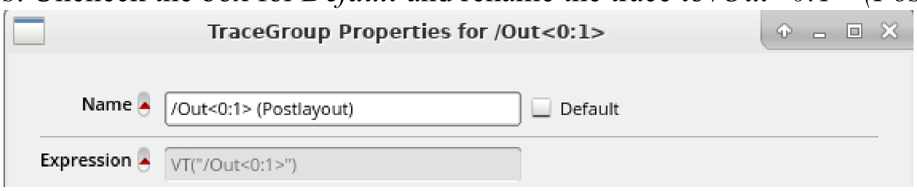


- Return to the *ADE Explorer* window. We need to tell ADE to use the config view instead of the schematic. To do this, select **Setup->Design.** Change *View Name* to *config*.

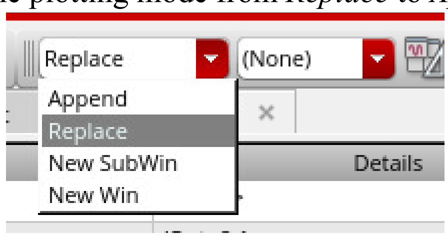
- The bottom right of the ADE window should now show that you are simulating the config view



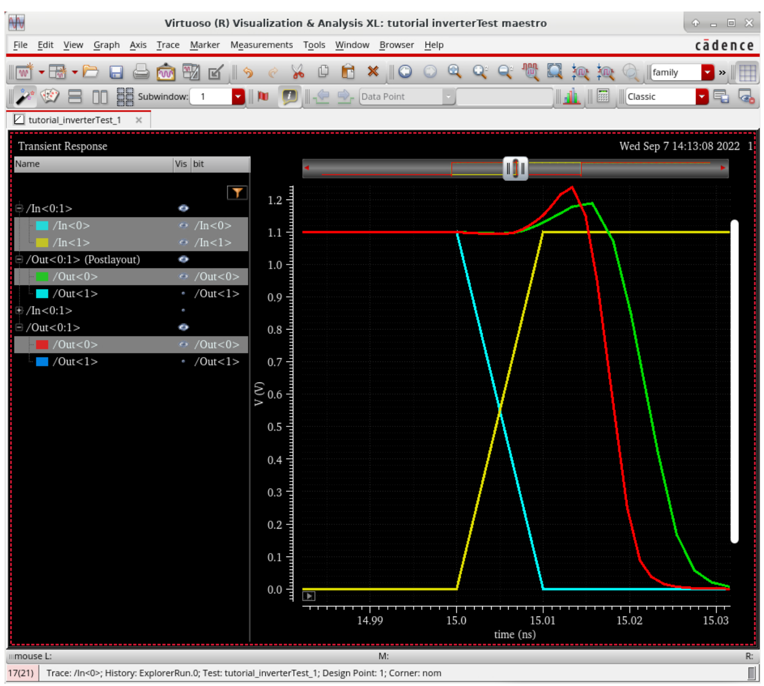
- Run the simulation. Right click on */Out<0:1>* in the plot window and select *TraceGroup Properties*. Uncheck the box for *Default* and rename the trace to */Out<0:1> (Postlayout)*.



- Return to the *Hierarchy Editor* window. Right click on inverter and set the view back to schematic then save the config.
- In *ADE Explorer*, change the plotting mode from *Replace* to *Append*



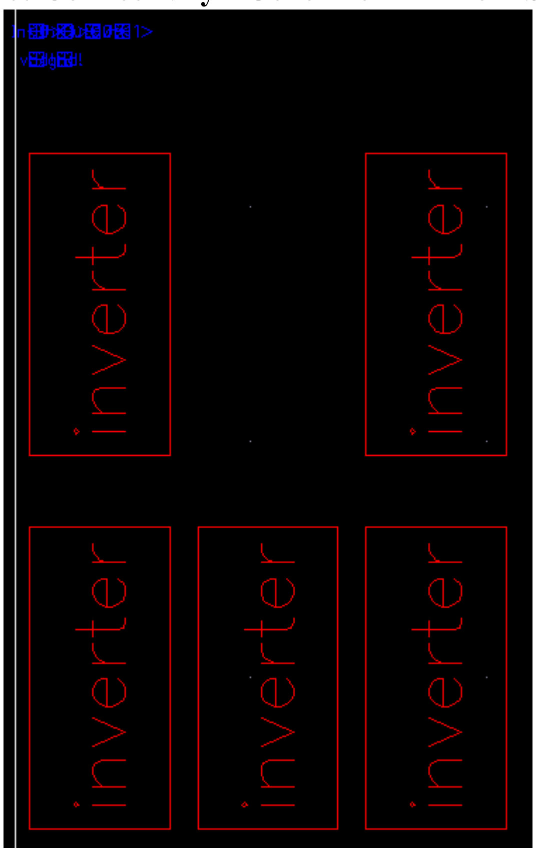
- Run the simulation. You can zoom in on a transition to compare the pre and postlayout performance.

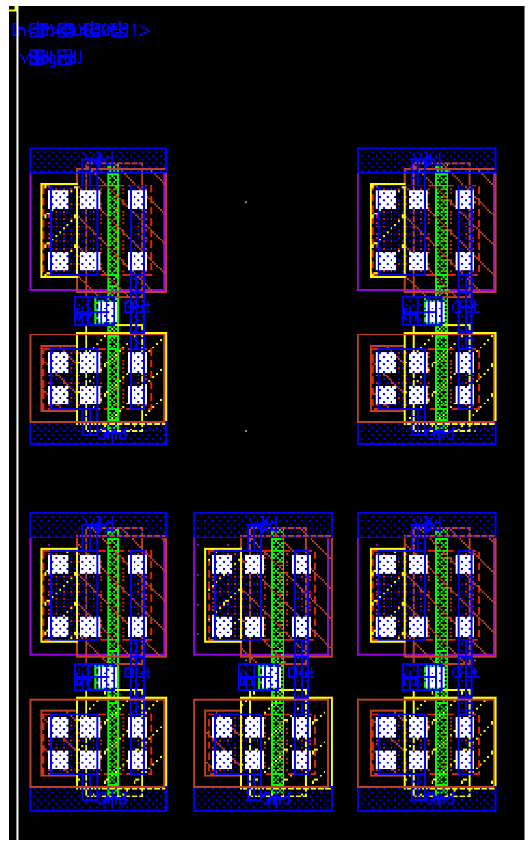


7 Hierarchal Layout Editing

Circuit designs typically consist of multiple instances of the same cell. An example would be the *InverterTest* design that you have created in the previous tutorial. Instead of having to redraw the same layout cellview multiple times in a higher level cellview, you can place completed layout instances in the design, similar to schematic editing. Since it is ensured that each of the cells function correctly by means of LVS checks, DRC, and Spectre simulation, placing layout instances not only saves time in drawing the cells, it means that the only errors that occur should result from connection errors.

- In the inverterTest schematic, select **Launch->Layout XL.** Create a new layout.
- Use **Connectivity->Generate->All from Source** to create the inverter instances and pins.





- By default, *Virtuoso* on renders objects drawn in the current cell. You can see the layout of all the hierarchical blocks by pressing **Shift + F**.
- You can return to showing only routing in the current cell by pressing **Ctrl + F**.

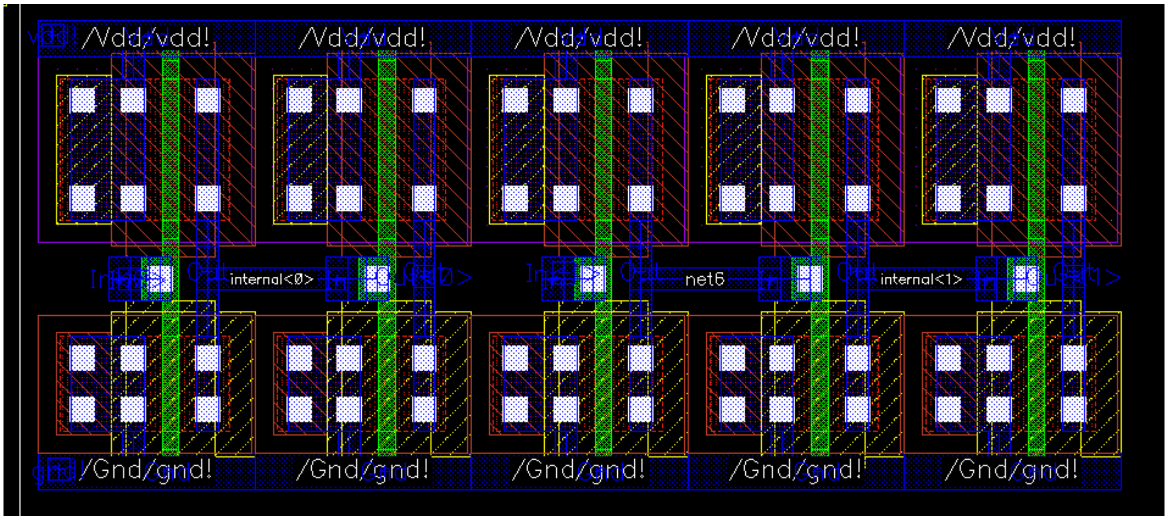
- Note that the layout editor does not allow you to edit the cell. However, if you edit one of the cells at the lower level, the other cells of the same instance in the higher level will be changed.

- Arrange the inverters in a row-based way that will allow you to use a single VDD and GND line. It will also lead to lower area.

- Keep only one VDD and one GND pin.

- Connect the inverters according to the *InverterTest* schematic.

- The finished layout should look like as in the next figure. Note that all of the PMOS devices have been placed so they are in the same Nwell. This would allow us to remove the bodyties from all but one inverter. Reducing the amount of bodyties will decrease the area used, but it also makes the design more susceptible to latchup.



- Perform Design Rule Check (DRC), Layout vs. Schematic (LVS) check, and Parasitic extraction using the same steps as the single inverter. Simulate your extracted layout to verify the expected functionality. The postlayout simulation of the complete inverterTest structure is the most representative of actual performance. Compare this performance to the simulation with the complete schematic design and the design with only postlayout simulation of the inverter cells.